

# Gan Minghao

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## EDUCATION

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| <b>The University of Hong Kong</b><br><i>Exchange Student in School of Computing and Data Science</i> | Jan. 2026 – May 2026<br>Hong Kong              |
| <b>Sun Yat-sen University</b><br><i>B.Eng. in School of Software Engineering</i>                      | Sep. 2023 – Present<br>Guangzhou/Zhuhai, China |
| – <b>Academic Records:</b> GPA 4.2/5.0, Rank: 10/231 (Top 5%)                                         |                                                |

## SCHOLARSHIP & AWARDS

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| <b>National Scholarship for Undergraduates</b> , Ministry of Education of China     | 2023 |
| <b>First Prize, Outstanding Student Scholarship</b> , Sun Yat-sen University        | 2023 |
| <b>Special Scholarship</b> , School of Software Engineering, Sun Yat-sen University | 2023 |
| <b>Second Prize, Outstanding Student Scholarship</b> , Sun Yat-sen University       | 2024 |
| <b>Fung Scholarship</b> , Fung Foundation & The University of Hong Kong             | 2025 |

## RESEARCH EXPERIENCE

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| <b>NAISS Lab, The University of Hong Kong</b><br><i>Research Assistant (Supervised by Prof. Shinan Liu)</i>                                                                                                    | Hong Kong<br>Jan. 2026 – Present          |
| – <b>Agent Infra Exploration</b>                                                                                                                                                                               |                                           |
| – <b>KV Cache Reuse across Architectures and Positions.</b> Investigated cross-architecture and positional KV cache reuse strategies to reduce redundant prefill computation in multi-turn agent interactions. |                                           |
| – <b>Agentic Serving Scheduling.</b> Explored scheduling optimizations targeting KV cache idle periods from tool-call latency to improve serving throughput.                                                   |                                           |
| – <b>Mixed-precision KV Cache inference backend.</b> Developing a general backend for mixed-precision KV Cache computation to support phase-aware KV Cache quantization in long-context agentic inference.     |                                           |
| <b>InPlus Lab, Sun Yat-sen University</b><br><i>Research Intern (Supervised by Prof. Wuhui Chen)</i>                                                                                                           | Guangzhou, China<br>Nov. 2024 – Dec. 2025 |
| – <b>Project: Low Latency LLM inference system on consumer-grade GPUs via Activation Sparsity</b>                                                                                                              |                                           |
| – Designed a bottleneck-aware neuron caching policy to maximize GPU workload in sparse computation.                                                                                                            |                                           |
| – Optimized a hybrid computing pipeline enabling asynchronous GPU&CPU execution with on-the-fly weights I/O.                                                                                                   |                                           |
| – Developed custom CUDA kernels for sparse matrix multiplication, optimizing hardware parallelism.                                                                                                             |                                           |
| – Achieved up to $3.7\times$ speedup in inference throughput over state-of-the-art systems; work accepted by <b>OSDI 2026</b> .                                                                                |                                           |

## PUBLICATIONS

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| <b>Kairox: Adaptive GPU-CPU Hybrid LLM Inference via Online Neuron Balancing</b>                         | <b>OSDI 2026</b> |
| Yapeng Jiang, <u>Minghao Gan</u> , Zicong Hong, Wuhui Chen, Junyuan Liang, Yue Yu, Meng Guo, Zibin Zheng |                  |

## SKILLS

**Programming** : C/C++, CUDA, Python  
**Frameworks** : Llama.cpp, GGML, vLLM, PyTorch  
**Tools** : NVIDIA Nsight Systems, NVTX, Git, SSH, Docker, LaTeX

**LANGUAGE PROFICIENCY**

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**IELTS Academic : 7.5** (Listening: 8.0, Reading: 8.5, Writing: 6.5, Speaking: 6.5)